Terrorism Dashboard User Manual

Purpose of the Dashboard

The **Terrorism Dashboard** is an interactive tool designed to:

- Explore incidents of terrorism across regions and countries.
- Understand patterns in **casualties, perpetrators, and targets**.
- Analyze **trends over time** using clear charts and maps.

Target Audience

This dashboard is useful for:

- **Students & Researchers** – academic projects or analysis.
- **Policy Makers & Analysts** – quick insights into trends and impact.
- **General Public** – anyone curious about terrorism data.

Before Running the Dashboard

Before opening the dashboard, make sure:

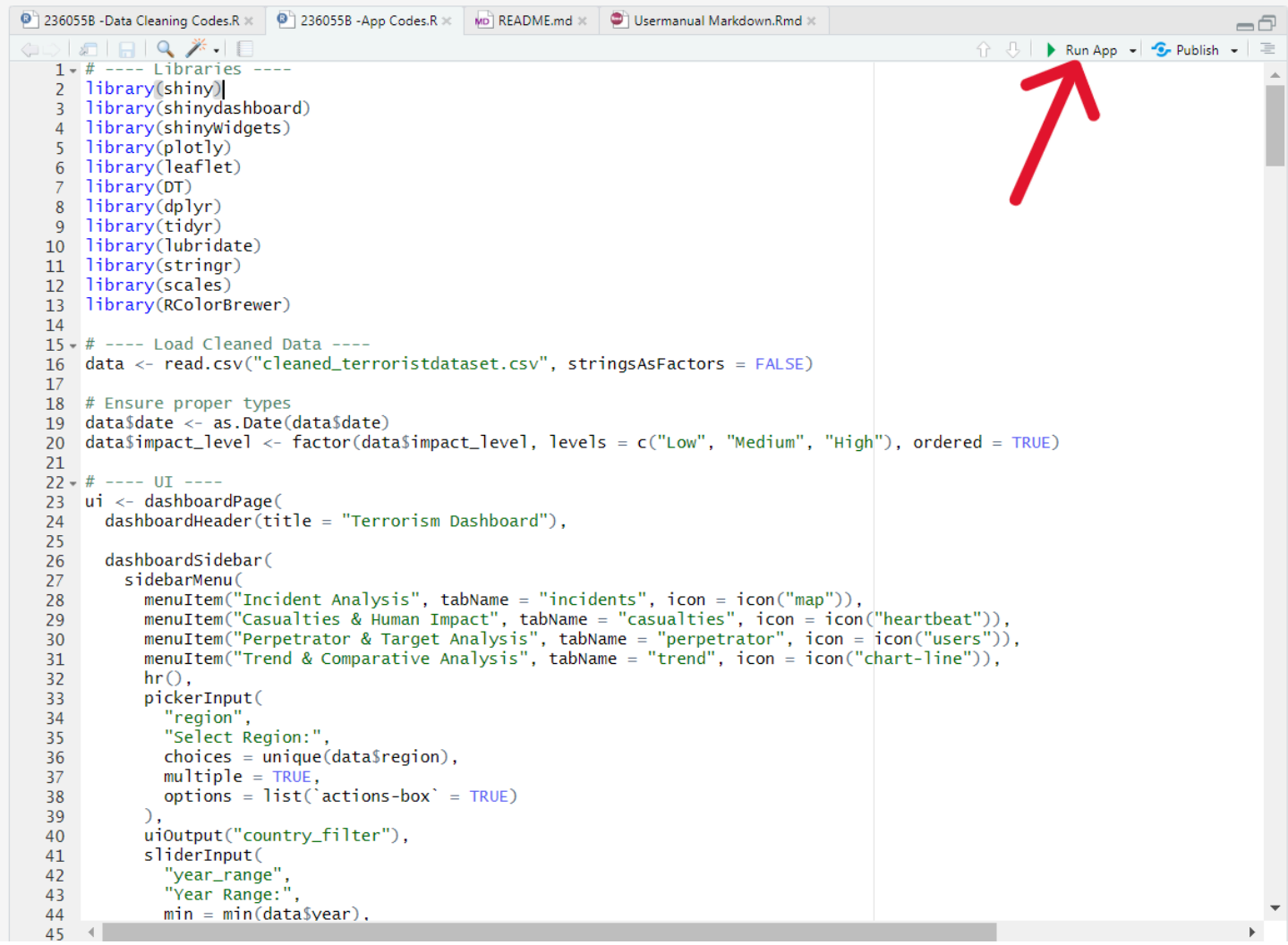
- **R and RStudio are installed** on your computer.
- **All required libraries are installed and loaded**. These include:
  - shiny
  - shinydashboard
  - shinyWidgets
  - plotly
  - leaflet
  - DT
  - dplyr
  - tidyr
  - lubridate

- stringr
- scales
- RColorBrewer

- **Dataset is available** – The cleaned terrorism dataset (236055B-cleaned_terroristdataset.csv) should be placed in the project folder.
- **Internet connection (optional)** – Needed if you plan to install packages during the first run.

Getting Started

1. **Open the Application**
   - Run the **236055B- App Codes. R** file in RStudio. Click on the Run App Icon to run the file.



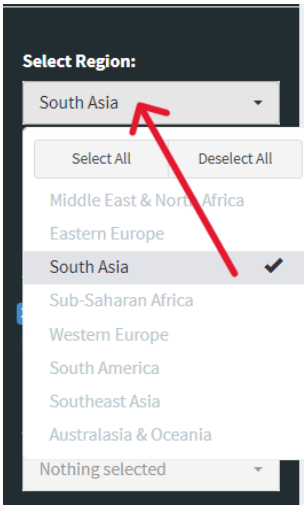
   - The dashboard will open in your browser.

Dashboard Structure

Filters

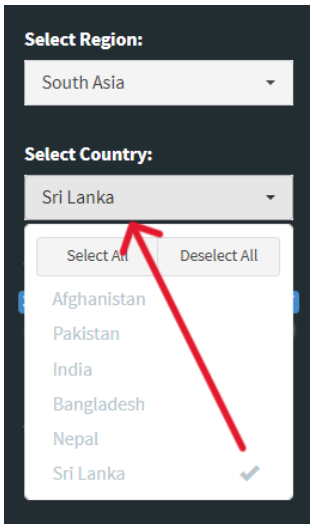
1. **Use the Sidebar Filters**
   - Region (choose one or multiple)

e.g : choosing South Asia



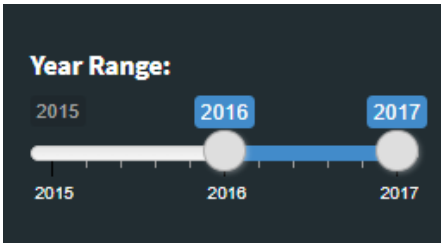
   - Country (updates based on region)

e.g: Selecting Sri Lanka as the country based on South Asia Region



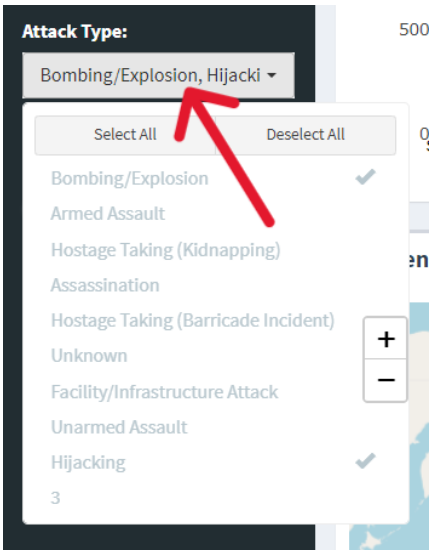
   - Year range (slider)

e.g : Choosing Year range as 2016- 2017



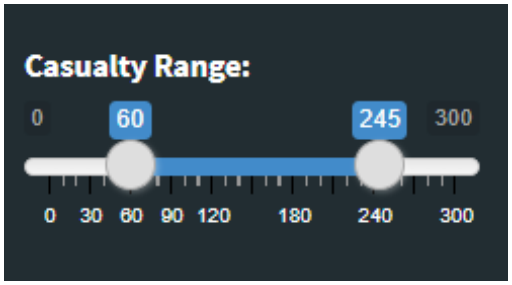
- Attack type (multi-select)

e. g: Selecting Bombing/Explosion & Hijacking as multiple selection



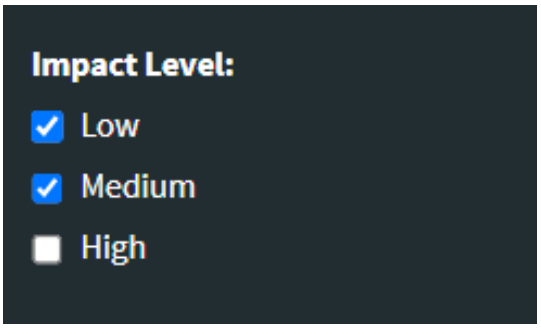
- Casualty range

e.g : Selecting Casualty range as 60 - 245



- Impact level (Low, Medium, High)

e. g: Selecting Impact level of both Low & Medium



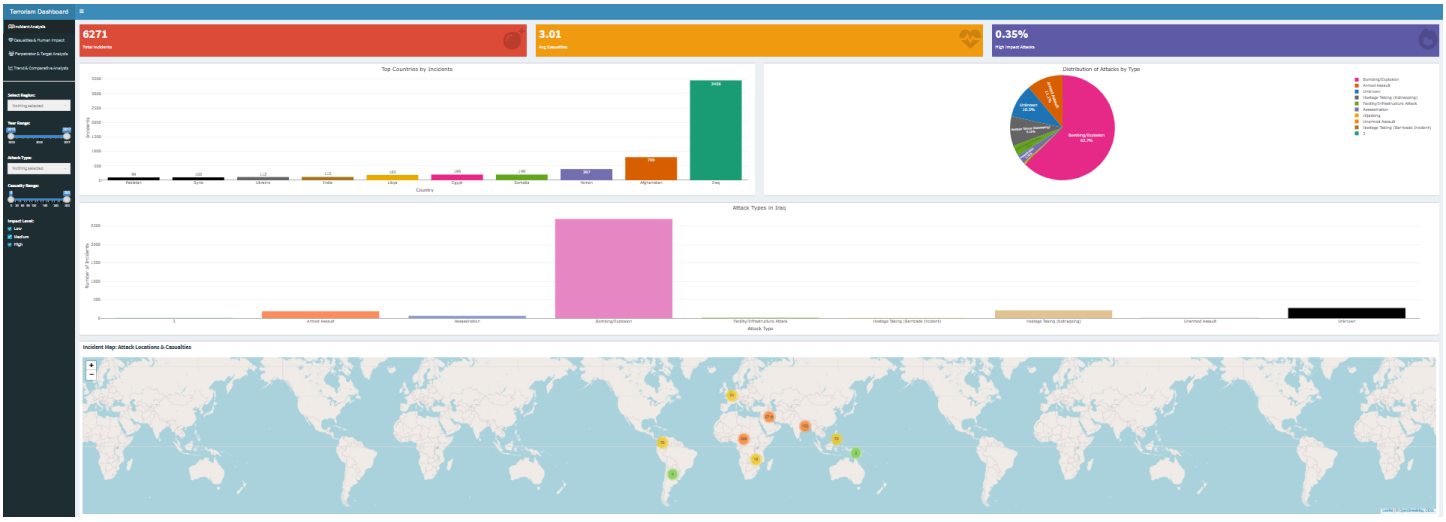
👉 These filters control what data is shown.

Dashboard Tabs

The dashboard has **four main sections (tabs):**

1. Incident Analysis Tab

- **KPIs:** total incidents, average casualties, % of high impact.
- **Visuals:**
  - **Top Countries (Bar Chart):** Shows which countries have the highest number of incidents.
  - **Attack Types (Pie Chart):** Displays the share of different attack types.
  - **Attack Types by Country (Drill-Down Chart):** Breaks down attack types for a selected country (the top country is shown by default).
  - **Interactive Map:** Explore incidents by location with zoom and hover options.



(Incident Analysis Tab)

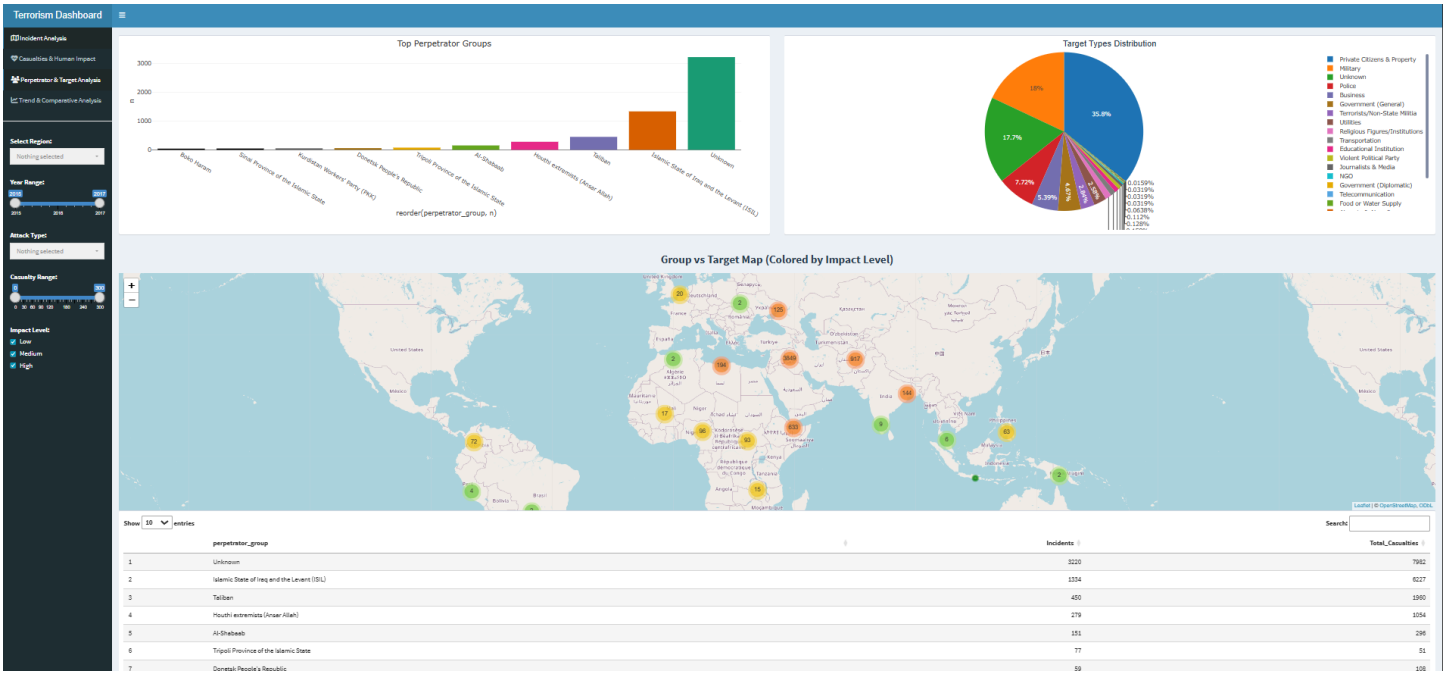
2. Casualties & Human Impact Tab

- **Killed vs Wounded (Line Chart):** Shows yearly trend of people killed vs wounded.
- **Casualties by Attack Type (Bar Chart):** Compares total casualties across different attack types.
- **Heatmap (Casualties by Region & Year):** Highlights regional casualty intensity over time.
- **Boxplot of Weapon Types:** Shows casualty distribution by weapon type (with code mapping table).
- **Total vs Avg Casualties by Region (Bar Chart):** Compares total casualties against regional averages.

(Casualties & Human Impact Analysis Tab)

3. Perpetrator & Target Analysis Tab

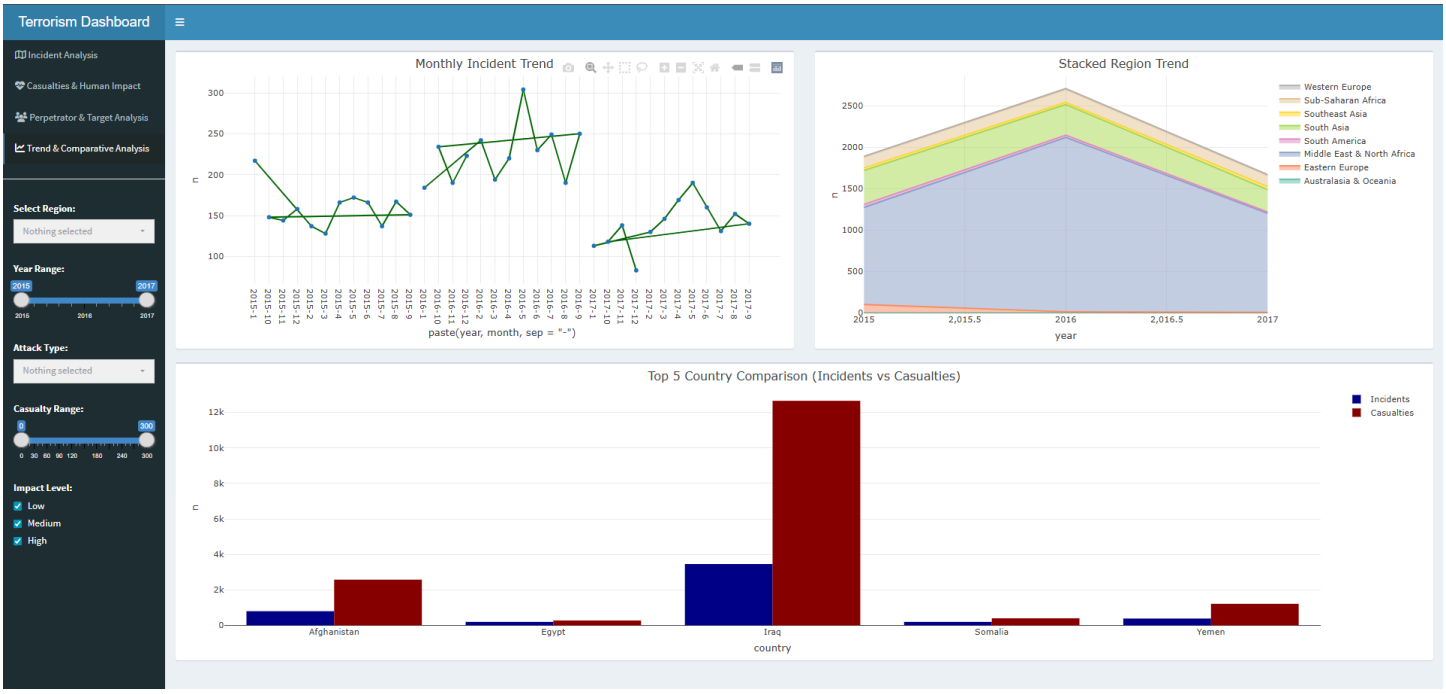
- **Top Perpetrator Groups (Bar Chart):** Displays the groups responsible for most incidents.
- **Target Type Distribution (Pie Chart):** Shows which types of targets were most attacked.
- **Group vs Target Map:** Interactive map linking perpetrators to their targets, colored by impact level.
- **Perpetrator Group Details (Table):** Summarizes number of incidents and casualties by group.



(Perpetrator & Target Analysis Tab)

4. Trend & Comparative Analysis Tab

- **Monthly Incident Trend (Line Chart):** Tracks number of incidents across months.
- **Stacked Region Trend (Area Chart):** Shows contribution of regions to total incidents over time.
- **Top 5 Country Comparison (Bar Chart):** Compares incidents vs casualties for the top 5 countries.

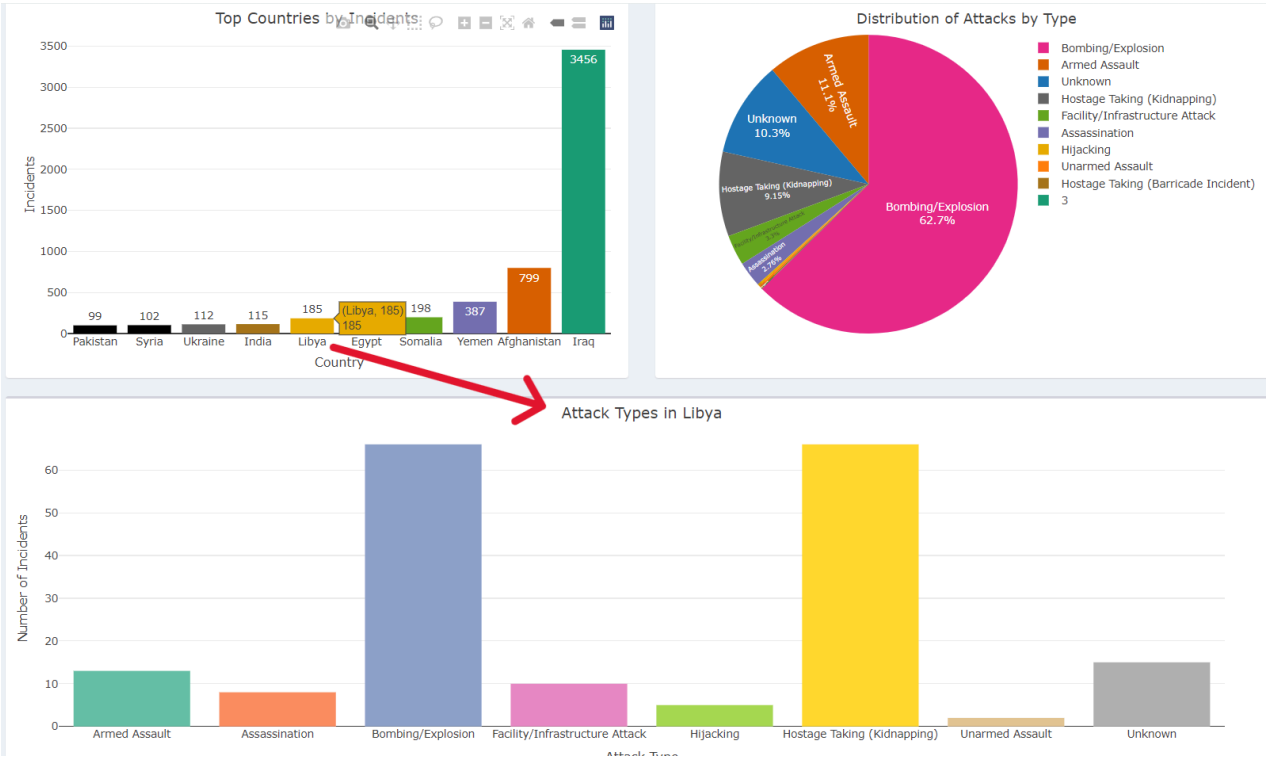


(Trend & Comparative Analysis Tab)

Interactive Features

- **Hover on charts** → see exact details.
- **Click legends** → show/hide categories.
- **Zoom & pan maps** → explore specific locations.
- **Sort tables** → click column headers.
- **Drill-Down** → View attack types inside a selected country for deeper insight.

e.g: click on the 'Libiya' to see the attack types in 'Libiya '



How to Read the Visualizations

- **Bar Charts:** Compare sizes across categories (e.g., top countries or attack types). Taller bars = higher values.
- **Pie Charts:** Show proportion of each category in relation to the whole. Larger slice = bigger share.
- **Line Charts:** Show trends over time (e.g., casualties per year). Rising line = increase, falling = decrease.
- **Area Charts:** Show cumulative trends with filled regions. Useful for comparing regions over time.
- **Heatmaps:** Darker colors = higher intensity (e.g., more casualties in that region/year).
- **Boxplots:** Show spread of data. The box = middle range, dots = outliers.
- **Maps:** Each marker or shaded area shows incidents in a location. Zoom and hover for more detail.
- **Tables:** Display detailed data (numbers, groups, summaries) in a structured way.

Troubleshooting

- **Dashboard not opening:**

  Make sure all required R packages are installed (shiny, shinydashboard, plotly, leaflet, DT, etc.).

  Use install.packages("package_name") if missing.

```r
# ---- Install Required Packages ----
# Run this code if you don't already have these libraries installed

required_packages <- c(
  "shiny", "shinydashboard", "shinyWidgets",
  "plotly", "leaflet", "DT",
  "dplyr", "tidyr", "lubridate",
  "stringr", "scales", "RColorBrewer"
)
```

- **Data not loading:**

  Confirm the dataset file (selected_terroristdataset.csv) is in the same folder as the dashboard files (app.R or ui.R + server.R).

  Check the file name matches exactly (including capitalization).

- **Charts not displaying:**

  Refresh the app in your browser (click the refresh button).

  Check if filters are set too narrowly (e.g., no data for chosen country/year). Reset filters to “All”.

- **Map not showing markers:**

  Ensure internet connection is active (map tiles load online).

  Try zooming out to see if markers are outside the current view.

- **App running slowly:**

  Close other heavy applications while running R/Shiny.

  Apply fewer filters or load smaller portions of the dataset.

- **Error messages in R console:**

  Read the message carefully — often it will point to the missing package, file, or syntax.

  Restart R session and re-run the app.

Thank you!

Contact for Support
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